

Topic 10: Organic Chemistry: Halogenoalkanes, Alcohols and Spectra

Related topics in Units 4 and 5 will assume knowledge of this material.

10A: General principles

Students will be assessed on their ability to:

10.1	be able to classify reactions (including those in Unit 1) as addition, elimination, substitution, oxidation, reduction, hydrolysis or polymerisation
10.2	understand the concept of a reaction mechanism
10.3	understand that heterolytic bond breaking results in species that are electrophiles or nucleophiles
10.4	know the definition of the term 'nucleophile'
10.5	understand the link between bond polarity and the type of reaction mechanism a compound will undergo

10B: Halogenoalkanes

Students will be assessed on their ability to:

10.6	understand the nomenclature of halogenoalkanes and be able to draw their structural, displayed and skeletal formulae
10.7	understand the distinction between primary, secondary and tertiary halogenoalkanes
10.8	understand the reactions of halogenoalkanes with: - i aqueous alkali, including KOH(aq) to produce alcohols (where the hydroxide ion acts as a nucleophile) - ii ethanolic potassium hydroxide to produce alkenes by an elimination reaction (where the hydroxide ion acts as a base) - iii aqueous silver nitrate in ethanol (where water acts as a nucleophile) - iv alcoholic ammonia under pressure to produce amines (where the ammonia acts as a nucleophile) - v alcoholic potassium cyanide to produce nitriles (where the cyanide ion acts as a nucleophile) <i>Students should know this is an example of increasing the length of the carbon chain.</i>
10.9	understand the mechanisms of the nucleophilic substitution reactions between primary halogenoalkanes and: - i aqueous potassium hydroxide - ii ammonia <i>S_N1 and S_N2 substitution mechanisms will be tested in Unit 4.</i>

10.10	understand that experimental observations and data can be used to compare the relative rates of hydrolysis of: i primary, secondary and tertiary structural isomers of a halogenoalkane ii primary chloro-, bromo- and iodoalkanes using aqueous silver nitrate in ethanol
10.11	CORE PRACTICAL 5 Investigation of the rates of hydrolysis of some halogenoalkanes.
10.12	know the trend in reactivity of primary, secondary and tertiary halogenoalkanes
10.13	understand, in terms of bond enthalpy, the trend in reactivity of chloro-, bromo- and iodoalkanes
10.14	CORE PRACTICAL 6 Chlorination of 2-methylpropan-2-ol with concentrated hydrochloric acid.
	Further suggested practicals: i the use of silver nitrate solution to identify the halogen present in halogenoalkanes ii preparation of 1-bromobutane from butan-1-ol, potassium bromide and sulfuric acid

10C: Alcohols

Students will be assessed on their ability to:

10.15	understand the nomenclature of alcohols and be able to draw their structural, displayed and skeletal formulae
10.16	understand the distinction between primary, secondary and tertiary alcohols
10.17	understand the reactions of alcohols with: i oxygen in air (combustion) ii halogenating agents • PCl_5 to produce chloroalkanes (including its use as a qualitative test for the presence of the $-\text{OH}$ group) • 50% concentrated sulfuric acid and potassium bromide to produce bromoalkanes • red phosphorus and iodine to produce iodoalkanes iii concentrated phosphoric acid to form alkenes by elimination <i>Descriptions of the mechanisms of these reactions are not required.</i>

10.18	understand that potassium dichromate(VI) in dilute sulfuric acid can oxidise: - i primary alcohols to produce aldehydes (which give a positive result with Benedict's or Fehling's solution) if the product is distilled as it forms - ii primary alcohols to produce carboxylic acids (which give a positive result with sodium carbonate or sodium hydrogencarbonate) if the reagents are heated under reflux - iii secondary alcohols to produce ketones <i>In equations, the oxidising agent can be represented by [O].</i>
10.19	understand, the following techniques in the preparation and purification of a liquid organic compound: - i heating under reflux - ii extraction with a solvent using a separating funnel - iii distillation - iv drying with an anhydrous salt - v boiling temperature determination
10.20	CORE PRACTICAL 7 The oxidation of propan-1-ol to produce propanal and propanoic acid.
	Further suggested practical: Investigation of reactions of primary and secondary alcohols, including propan-1-ol and propan-2-ol

10D: Mass spectra and IR

Students will be assessed on their ability to:

10.21	be able to interpret data from mass spectra to suggest possible structures of simple organic compounds using the m/z of the molecular ion and fragmentation patterns
10.22	be able to use infrared spectra, or data from infrared spectra, to deduce functional groups present in organic compounds, and predict infrared absorptions, given wavenumber data, due to familiar functional groups including: - i C–H stretching absorptions in alkanes, alkenes and aldehydes - ii C=C stretching absorption in alkenes - iii O–H stretching absorptions in alcohols and carboxylic acids - iv C=O stretching absorptions in aldehydes, ketones and carboxylic acids - v C–X stretching absorption in halogenoalkanes - vi N–H stretching absorption in amines
10.23	CORE PRACTICAL 8 Analysis of some inorganic and organic unknowns.